Ramit Pahwa

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Education

- **University of Texas at Austin** — Austin, TX
  M.S. in Computer Science (Multimodal CV and NLP) — *Aug 2022 – Apr 2024*
  - Research Assistant in **L.I.V.E Lab**, advised by **Prof. Alan Bovik**, working on Video Quality Assessment.
- **Indian Institute of Technology, Kharagpur (IIT KGP)** — Kharagpur, India
  Integrated M.S./B.S. in Mathematics and Computing; GPA: 9.0/10.0 (Salutatorian) — *Aug 2014 – May 2019*

Work Experience

- **Rivian and Volkswagen Tech** — Palo Alto, CA
  Senior Research Engineer — Speech, Multimodal AI & Post-Training — *May 2024 – Present*
  - **Omni Foundation Model (30B Post-Training)**: Tech Lead post-training a proprietary **30B** Omni model for edge spoken dialogue: SFT + preference optimization (DPO/RLHF-style) for multi-turn voice; task-token vocabulary extension improving instruction following and reducing hallucination.
  - **Full-Duplex Speech Model (PersonaPlex)**: Fine-tuned NVIDIA's **PersonaPlex** for native tool calling via streaming ASR + SLM tool router; **GRPO** post-training for latency-masking dialogue; added in-vehicle skills (navigation, messaging) as native capabilities.
  - **Voice-LM Evaluation (VLME)**: Designed **VLME**, a benchmark quantifying paralinguistic metadata in voice-language model evaluation with an adversarial gating protocol. Submitted to **PALM @ NeurIPS 2026**.
  - **Multilingual ASR**: Training a **216M** multilingual streaming ASR model **from scratch**; RL finetuning (GRPO, MWER-style lattice RL) and pruning + distillation compressing ASR models $\sim 3\times$ for on-device deployment.
  - **Speech Tool Calling**: First author on **Audio2Tool**, a large-scale audio-to-tool-call dataset and benchmark; built data synthesis pipeline and seq2seq models with SOTA intent-to-function mapping. Submitted to **Interspeech 2026**.
  - **Wake Word Detection**: Tech Lead (team of 4) for on-device wake word detection in Rivian vehicles: streaming KWS architecture, noise-robust training with hard-negative mining. **Contributed to Live-Kit.Filed 4 patents.**
  - **Voice Model (TTS)**: Led production TTS (CosyVoice, Qwen3-TTS, VITS) for English/German; distillation pipeline compressing TTS models to edge size with <1% MOS degradation.
  - **Multimodal Alignment & Data Flywheel**: Fine-tuned Qwen2.5-VL with RLHF-style reward shaping for video tagging (10k+ clips/day, **1 patent**); built active-learning package curating 50k hours of in-the-wild audio/video.
  - **Generative & Restoration Models**: Language-guided weather restoration; diffusion-based 2D character animation from text prompts — **Best Design Award, patent filed**.
- **Secta.ai** — Austin, TX
  Machine Learning Scientist: Generative AI, Diffusion Models, LLM, CLIP, VectorDB — *Aug 2023 – May 2024*
  - **LLM-Based Image Quality Assessment**: Built LLM reward models for image quality rating; curated preference data and fine-tuned with RLHF for enterprise image generation.
- **Adobe** — San Jose, USA
  Software Engineer (ML) — *May 2018 – Jun 2022*
  - **ML Platform in Acrobat**: Built streaming inference pipeline for Liquid Mode; shrank on-device models for Android; trained PDF summarization models and a lightweight conversational agent.
  - **Knowledge Distillation**: RL-guided CNN layer pruning with no accuracy loss — Data-Driven Compression of CNNs (ACM'20); distilled RNNs into CNNs for fast text classification — Model Blending (thesis).
  - **Mobile Platform**: Led MDM/MAM compliance library for Acrobat/Scan Android (**Tech Summit 2022**); built frame-profiling service establishing app performance baselines.

Publications

- **VLME: Paralinguistic Metadata in Voice-Language Model Evaluation**
  Ramit Pahwa et al. — *PALM @ NeurIPS'26*
- **Audio2Tool: Bridging Spoken Language Understanding and Function Calling [Paper]**
  Ramit Pahwa et al. — *Interspeech'26*
- **ASL-LIVE: Quality Assessment of American Sign Language Videos**
  Ramit Pahwa et al. — *TIP'26*
- **Perceptual Video Quality Assessment: The Journey Continues! [Paper]**
  Avinab Saha, Ramit Pahwa* et al.* — *Frontiers'23*
- **Data-Driven Compression of Convolutional Neural Networks [Code] [Paper]**
  Ramit Pahwa et al. — *ACM'20*
- **LSTMs with Attention for Aggression Detection [Code] [Paper]**
  Nishant Nikhil, Ramit Pahwa et al. — *COLING'18*

Patents

- **Wake Word Detection** 2025
  Ramit Pahwa et al. *US PTO Filed*
- **Two Stage Wake Word Detection** 2025
  Ramit Pahwa et al. *US PTO Filed*
- **Contextual Cabin Noise Suppression** 2025
  Ramit Pahwa et al. *US PTO Filed*
- **Diffusion-Based 2D Animation Creation Using Text Prompts** 2023
  Ramit Pahwa et al. *US PTO Approved*
- **Transformers for In-Application Text-Based Reading** 2021
  Ramit Pahwa et al. *US PTO Approved*